

CUSTOMER ANALYTICS • ML CLASSIFICATION

Customer Churn Prediction

Persevex Data Science Internship

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Project Overview

CHURN

Customers who stop using a service or product.

The goal

**Identify at-risk customers early —
then act before they leave.**



Problem Statement & Objectives

How might we identify customers likely to leave — in time for a meaningful retention action?

01

Analyze behavior

02

Find churn patterns

03

Build classifiers

04

Predict probability

05

Assign risk tiers

06

Recommend actions

Dataset & Features

3,000
customer records

27
model features

**Customer
information**

**Plan & billing
frequency**

**Login & usage
activity**

**Payment
delays**

**Support &
resolution**

**Engagement
indicators**

Data Preprocessing & Feature Engineering



80 : 20 STRATIFIED • STANDARD SCALING • ONE-HOT ENCODING

Engineered signals

Expected Login 7D	Login Drop %	Expected Usage 7D
Usage Drop %	Support Friction	Engagement Score
High Inactivity	Payment Risk	Low Engagement

Machine Learning Models

Logistic Regression

Interpretable linear baseline for churn probability.

Random Forest

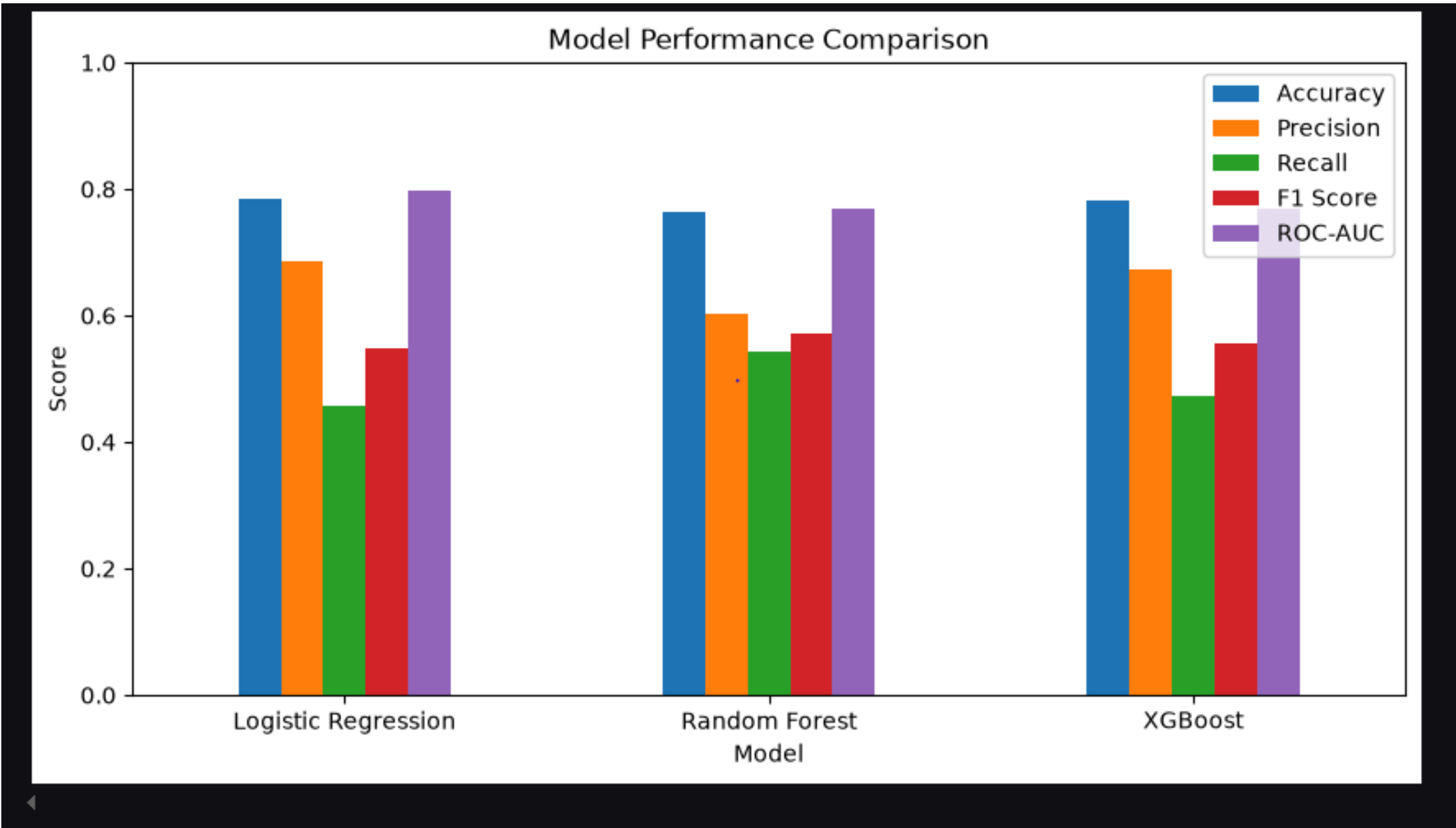
Ensemble of decision trees capturing nonlinear interactions.

XGBoost

Boosted trees optimized sequentially for difficult cases.

TRAINING SETUP 80:20 SPLIT • STRATIFIED • `random_state = 42`

Model Evaluation

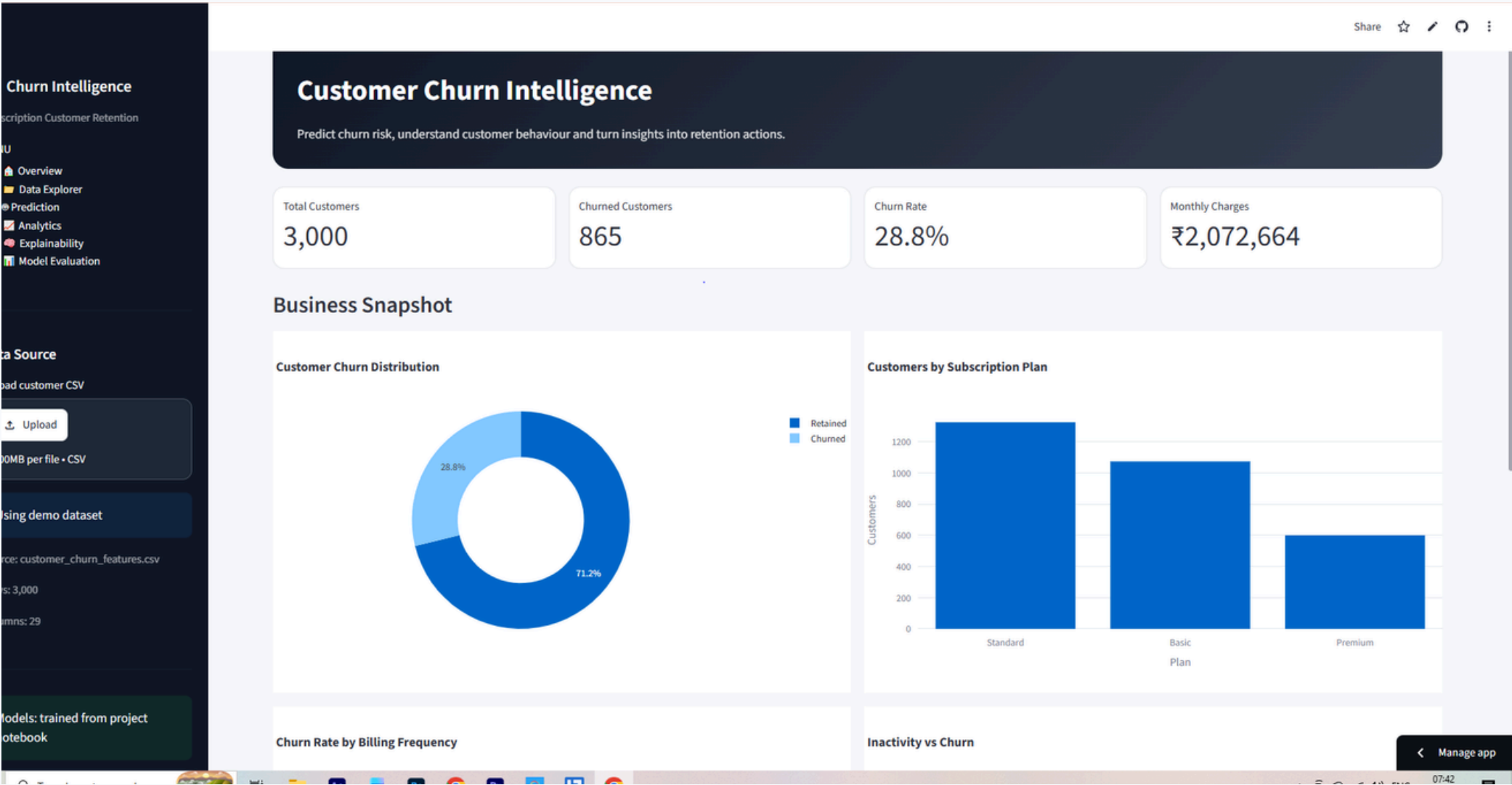


COMPARISON LENS

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

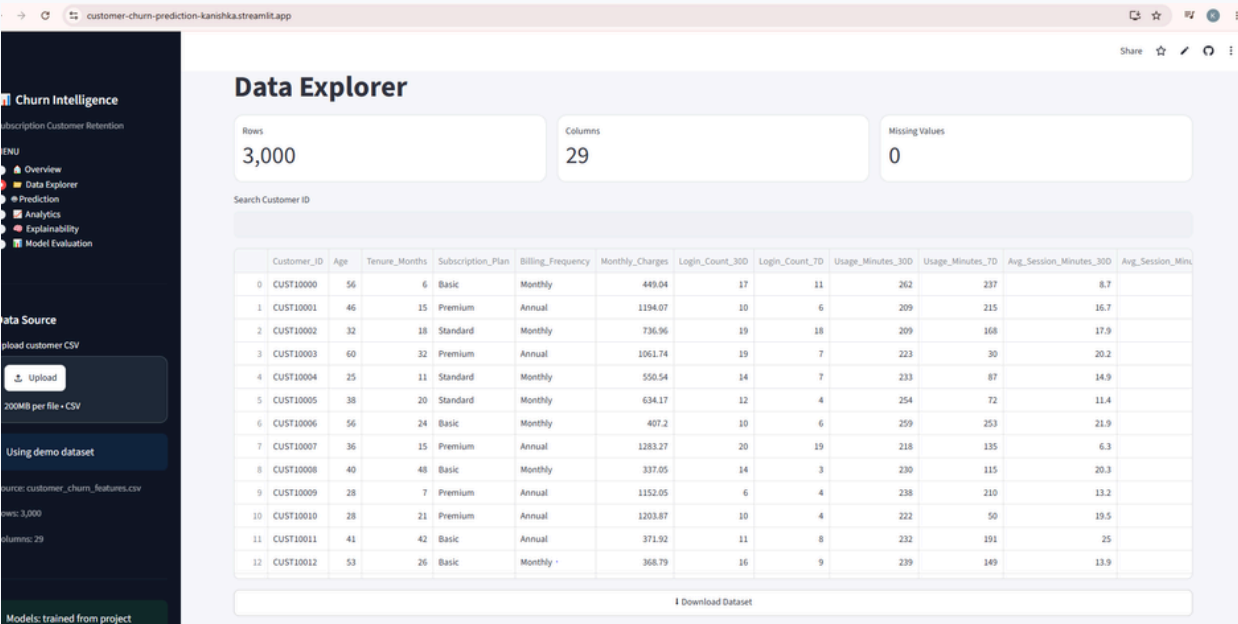
PR-AUC was also reviewed, but is not shown in the supplied chart. Model selection compares the visible metrics without assuming unsupported values.

Customer Churn Prediction Dashboard



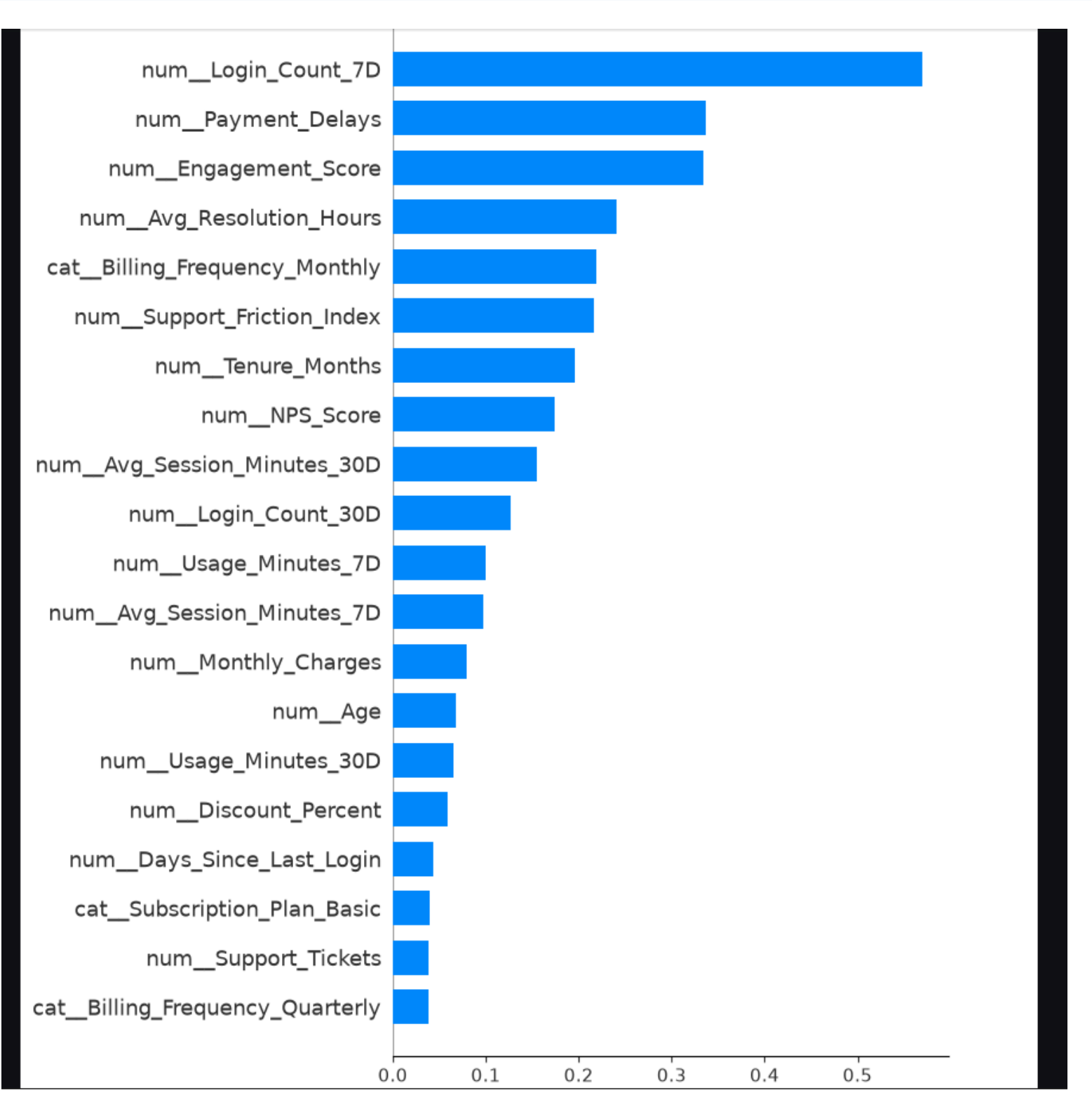
01 OVERVIEW

Churn stats • Interactive prediction • Probability & risk tier • Analytics • Downloadable results



02 DATA EXPLORER

Explainability & SHAP Analysis



Trust needs explanation.

SHAP values show how each feature influences model output — turning a probability into an interpretable business signal.

KEY INFLUENCES

Login counts • Payment delays
Engagement score • Billing frequency
Support friction • Tenure • NPS

Prediction & Risk Classification

Share ☆ ↻ 🔍 ⋮

Churn Prediction

The models are trained automatically with the exact configuration from the model-training notebook, avoiding saved-pickle version conflicts.

Select Model

Logistic Regression

Churn probability threshold

0.50

	Customer_ID	Age	Tenure_Months	Subscription_Plan	Billing_Frequency	Monthly_Charges	Login_Count_30D	Login_Count_7D	Usage_Minutes_30D	Usage_Minutes_7D	Avg_Session_Minutes_30D	Avg_Session_Minutes_7D
0	CUST10000	56	6	Basic	Monthly	449.04	17	11	262	237	8.7	
1	CUST10001	46	15	Premium	Annual	1194.07	10	6	209	215	16.7	
2	CUST10002	32	18	Standard	Monthly	736.96	19	18	209	168	17.9	
3	CUST10003	60	32	Premium	Annual	1061.74	19	7	223	30	20.2	
4	CUST10004	25	11	Standard	Monthly	550.54	14	7	233	87	14.9	
5	CUST10005	38	20	Standard	Monthly	634.17	12	4	254	72	11.4	
6	CUST10006	56	24	Basic	Monthly	407.2	10	6	259	253	21.9	
7	CUST10007	36	15	Premium	Annual	1283.27	20	19	218	135	6.3	
8	CUST10008	40	48	Basic	Monthly	337.05	14	3	230	115	20.3	
9	CUST10009	28	7	Premium	Annual	1152.05	6	4	238	210	13.2	

🚀 Run Churn Prediction



FROM SCORE TO ACTION

Predicted churn vs. retained
Probability score per customer
High / Medium / Low risk tiers
Download results as CSV

Business Insights & Retention Strategy

01

Prioritize

Identify high-risk customers for intervention.

02

Monitor

Watch declining login and usage behavior.

03

Resolve

Investigate payment delays and support friction.

04

Engage

Trigger timely, relevant retention actions.

Prediction creates value only when it changes the next customer interaction.

Machine learning turns behavior into an early retention signal.

NEXT

- Real-time prediction integration
- Model monitoring and updates
- Personalized retention recommendations
- Larger, continuously updated datasets
- Deployment into business systems

Thank You

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